Def Agent_ benchmarks( ): #4

⇒ Consider tasks that require tool use (e.g., running code) and iterating over a period of time

⇒ Agent includes: LLM, memorization, tool, workflow

⇒ Agent = language model + agent scaffolding

(logic for deciding how to use the LM)

⇓ SWE Bench:
- 2294 tasks across 12 python repo.
- Given codebase + issue description ⇒ submit a PR.
- Evaluation metric: Unit test.

paper: SWE Bench: Can language Models Resolve Real-world Github issues? : Jimenez, et al., -- 2024

⇓ CyBench: A Framework for Evaluating Cybersecurity capabilities and Risks of language models : Zhang + 2024

↳ includes 40 capture the Flag (CTF) tasks
use first-Resolve time as a measure of difficulty

⇓ MLE Bench: 75 Kaggle competitions: requiring processing data, train/test model.

↳ competition description → Dataset leaderboard → Agent (process data, Train model, test model, debug, create submission) → submission → grader → score: 63.4%

Def pure-Reasoning_benchmarks(): #5

All tasks require linguistic and world knowledge

Can we isolate reasoning from knowledge?

Arguably, reasoning captures a more pure form of intelligence

(isn't just about memorizing facts)

⇓ ARC-AGI-1
ARC-AGI-2 : François Chollet: creator of keras lib.

(Abstract-Reasoning corpus - Artificial General intellegence)

AGI: is not only a system that can automate most of economically valuable work,

but also a system that can automatically acquire new skills /knowledges outside of its training data

---

safety ~~safe~~ ~~safety~~

Def ~~safe~~ safety_benchmark(): #6

what does safety mean for AI?

(HELM safety: curated set of benchmarks): stanford (HAI) website

⇓ HarmBench: Based on 510 harmful behaviors that violate laws or concerns

⇓ AIR-Bench: Based on regulatory frameworks and company polices
Taxonomized into 314 risk categories, 5694 prompts

⇓ jailbreaking: LLM Refuses harmful instructions after trained
⇒ Greedy Coordinate Gradient automatically opimizes prompts to bypass safety
⇒ Transfer from open-weight models (LLama) to closed models (GPT-4)